



Material Safety Data Sheet

HEIDRIC INDUSTRIAL GAS TRADING

Trade Name: Refrigeration Grade Ammonia

Section 1	Chemical Product and Company Information
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Manufacturer or Formulator:	Heidric Industrial Gas Trading Barangay Sta. Catalina Sur Candelaria, Quezon Province	Product Name: Refrigeration Grade Ammonia Common Name: Ammonia, NH ₃ , Anhydrous Ammonia Chemical Type: Alkaline Gas
Email:	HeidricIGT.ops@gmail.com	
Mobile:	+63 949 363 8688 +63 919 998 8580	

Section 2	Composition/Information on Ingredients
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Chemical Name & Synonyms	C.A.S. No.	Chemical Formula	WT% Hazardous	TLV	PEL
Ammonia, Anhydrous Ammonia Liquefied Ammonia	7664-41-7	NH ₃	99.995	25 ppm 18 mg/cum 35 ppm STEL 50 ppm CI	50 ppm 35 mg/M
Inert Ingredients			Non-Hazardous .005		

Section 3	Hazard Identification
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Ingestion:	Ingestion of liquid ammonia may result in severe irritation or ulceration of the mouth, throat, and digestive tract which may be displayed by nausea, vomiting, diarrhea, and in severe case, collapse, shock and death. Unrecognized late strictures may be confused with malignant growths.
Inhalation:	Ammonia vapors cause irritation of the eyes and respiratory tract. High concentrations cause conjunctivitis, laryngitis and pulmonary edema, possibly accompanied by a feeling of suffocation. Causes secretion of saliva and retention of urine. A single breath of intensive at 5,000 ppm may cause serious adverse effects, lung injury or even death.
Eye Contact:	Exposure to high gas concentrations may cause temporary blindness and severe eye damage. Direct contact to the eyes with liquid ammonia will produce serious eye burns. Ammonia has a greater tendency than other alkali to penetrate and damage the eye, and to cause cataracts. (EPA, 1986) Iritis may be accompanied by hyppoyon or hemorrhages, extensive loss of pigment and severe glaucoma. High concentrations may result in permanent impairment of vision, even blindness.
Skin Absorption:	Ammonia gas may cause skin irritation, especially if skin is moist. The liquid can cause damage, resulting from combined freezing and corrosive action on the skin. If absorption becomes extensive, coma may arise preceded by convulsion.
Skin Contact:	Liquid ammonia produces severe skin burns on contact. Atmospheric concentration above 30,000 ppm will burn and blister skin after a few seconds of exposure.
Effect of Overdose:	Irritation and burns of skin and mucous membranes, headache, salivation, nausea, and vomiting. Dyspnea and cough with bloody mucous discharge. Bronchitis, laryngitis, hemoptysis, and pulmonary edema or pneumonitis. Death may result. Damage to the eyes may be permanent.

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Section 4	First Aid Measure
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Inhalation: REMOVE FROM CONTAMINATED AREA AND GET MEDICAL HELP IMMEDIATELY
Ingestion: If the patient is conscious, he should try to drink large quantities of water or weak acids such as dilute vinegar or citrus juice. NEVER GIVE FLUIDS OR INDUCE VOMITING IF PATIENT IS UNCONCIOUS OR HAVING CONVULSIONS.

Inhalation: In severe exposure, if needed, apply artificial resuscitation. Call or take to a doctor immediately. Prompt action is necessary.

Eyes: Immediately flush with flowing water for at least 15 minutes with the eyelids held apart. No oil or other non-water soluble preparation should be placed into the eyes. TAKE TO A DOCTOR.

Skin: Promptly flush the skin with flowing water for at least 15 minutes. IMMEDIATELY REMOVE contaminated clothing, being careful not to tear the skin. Do not use salves or ointment on skin for mucous membrane burns for 24 hours. Treat as for thermal burns.

Exposure: IN ALL CASES CALL OR TAKE TO A DOCTOR IMMEDIATELY!

Section 5	Fire Fighting Measure
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Extinguishing Media: Water spray or fog streams (chemical of CO2 on small fires only)

Special Fire Fighting Procedures: Stop flow of gas. Use water fog to keep fire-exposed containers cool and to protect person effecting the shutoff. Wear self-contained breathing apparatus. Wear full protective clothing. Approach fire upwind and evacuate area downwind.

Unusual Fire and Explosion Hazards: The presence of oil or other combustible materials will increase the fire hazard. Moderate when exposed to flame. Forms explosive compounds in contact with Ag, acetaldehyde, acrolein, B, BI3, halogens, BrF5, HClO3, ClO, ClF3, Chlorites, chlorosilane, CrO3, chromyl chloride, (ethylene dichloride + liquid ammonia), ethylene oxide, Au, hexachloromelamine, (hydrazine + alkali metals), HBr, HOCl, Mg(ClO4)2, Hg, HNO3, NO2, N2O4, NCl3, NF3, nitryl chloride, OF2, P2O3, picric acid, (K + AsH3), (K + PH3), (K + NaNO2), KClO3, potassium ferricyanide, potassium mercuric cyanide, AgCl (K+CO), S, SCI2, tellurium hydropentachloride, trichloromelamine.

Section 6	Accidental Release Measure
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Environmental Precautions: Highly toxic to fish and aquatic life. Do not contaminate any body of water by direct application, cleaning of equipment, or disposal. Since ammonia is a fertilizer, it may promote algae growth in waterways.

Steps to be taken in case material is released or spilled: Do not apply water onto leaking tank as water will raise the temperature and increase the leakage of the gas. Stop the flow of gas or liquid. Use water fog to protect persons effecting the shut-off. Wear full protective clothing and self-contained breathing apparatus. Approach upwind. Dike liquid spills to contain liquid. Evacuate the area immediately. Eliminate all open flames in vicinity of indoor spills or released vapor. Water fog can be used to cleanse atmosphere of ammonia vapor. Water fog nozzles positioned downwind can protect downwind areas.

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Section 7	Handling and Storage
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Precautions to be taken in handling and storing:

Protect against physical damage. Do not cut, heat, or weld on containers. Outside or detached storage is preferred. Inside storage should be cool, dry, well-ventilated noncombustible location, away from all possible source of ignition. Separate from food and feedstuffs. Separate from other chemicals, particularly oxidizing gases, chlorine, bromine, iodine, and acids. Avoid ingestion, inhalation, and contact with skin or eyes. Unsuitable storage includes copper or aluminum and their alloys, bases bronze, and galvanized surfaces, phenolic or polyester resins, and poly vinyl chloride resins. Do not use brazed joints for ammonia service.

Section 8	Exposure Controls/Personal Protection
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Ventilation Protection: Local exhaust is essential. Spark-proof fans with mechanical ventilation. Ducts should be located ceiling level and lead upward to the outside.

Respiratory Protection: Respiratory protection should be as follows: up to 300 ppm: any chemical cartridge respirator with cartridge(s) providing protection against the compound concern; any air-purifying full facepiece respirator (gas mask) with chin-style or front- or back-mounted canister providing protection against the compound concern; or self-contained breathing apparatus. Emergency or planned entry in unknown concentration or IDLH conditions: any self-contained breathing apparatus with positive pressure mode or any supplied-air respirator with full facepiece operated in pressure-demand or other positive pressure mode in combination with an auxiliary self-contained breathing apparatus and operated in pressure-demand or other positive pressure mode. Escape: any air purifying full facepiece respirator (gas mask) with a chin-style or front- or back-mounted canister providing protection against the compound concern or any appropriate escape-type self contained breathing apparatus.

Protective Clothing: Chemical protective clothing, (boots, protective gloves, and gas-tight goggles), composed of butyl rubber, neoprene, nitrile rubber and polyvinyl chloride. Viton is not recommended for use.

Suit Material Performance: (suggested material by E.P.A. – user should determine by specific use)

Butyl/Neoprene.....Good
Neoprene..... Good
PVC.....Good
Vitron.....Poor

Eye Protection: Gas tight chemical safety goggles must be worn when working with ammonia.

Other: Eyewash fountain and safety shower available in area. Person subject to ammonia exposure must not wear contact lenses.

Section 9	Physical and Chemical Properties
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Boiling Point:	-33.4 °C	Solubility in Water:	89.9/100 (0°C)
Density:	Sp. Gr. @ -33.4 °C = 0.68		7.4g/100cc (100°C)
Flash Point:	N/A, substance is Flammable	% Volatiles (by volume):	100
pH:	13+	Vapor pressure, mm Hg:	92.9 psig @ 15.6 °C
Appearance:	Colorless gas or liquid with extreme pungent odor		271.7 psig @ 48.9 °C
Extinguishing Media:	Water spray or fog type streams	Reaction with Water:	Exothermic

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Section 10	Stability and Reactivity
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Stability (Normal Conditions): Stable
Conditions to Avoid: Heat, open flames and electrical equipment and fixtures which are not vapor-proof or grounded. Critical temperature = 133°C
Incompatibility (Materials to avoid): Contact with oxidizing gases, chlorine, bromine, iodine, and acids. Mercury, silver oxide or hypochlorite can form explosive compound.
Hazardous Decomposition Products: Combustion will yield nitrogen and water.
Hazardous Polymerization: Will not occur.
Conditions to Avoid: Not applicable.

Section 11	Toxicology Information
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Ingestion: Toxicity Oral rat LD50: 350mg/kg
Inhalation: Lowest lethal concentration for human; ihi-hmm LCLo: 3000 ppm/5minutes; ihi-hmn LCLo: 5000 ppm/5minutes; ihi-rat LCLo: 2000 ppm/4hours. Inhalation mouse LC50: 4230 ppm/1hour. Ihi-cat and rabbit LCLo: 7000 ppm/1 hour.

Aquatic Toxicity Ratings: TLm 96: 10-1 ppm

Section 12	Ecological Information
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The substance is very toxic to aquatic organism.

Section 13	Disposal Considerations
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Waste Disposal Procedure:
Allow spilled liquid ammonia to evaporate, as it is too volatile to absorb. Diking will contain the liquid and allow it to stabilize. Keep unprotected personnel away from area until it is free of ammonia. Do not apply water directly to ammonia liquid.

Section 14	Transport Information
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Shipping Name: Ammonia, Anhydrous Ammonia, 2.3, UN 1005, Inhalation Hazard, stcc 49-042-10
UN Hazard Class: 2.3 C.A.S. Number: 7664-41-7
Reportable Quantity: 100lbs/18 gal D.O.T. Number: UN1005
Label Required: Flammable, Toxic Haz Waste No: D003

Refer to ICSC: 0414 Hazardous Materials Table for further provisions, packaging and quantity limitations.

Trade Name: Refrigeration Grade Anhydrous Ammonia Registration No: None
Placard: Non-Flammable EPA Regist No: None

Section 15	Regulatory Information
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U.S. FEDERAL REGULATIONS: EPA - ENVIRONMENTAL PROTECTION AGENCY CERCLA: Comprehensive Environmental Response, Compensation, and Liability Act of 1980 (40 CFR Parts 117 and 302) Reportable Quantity (RQ): 100 lbs (45.4 kgs)
SARA TITLE III: Superfund Amendment and Reauthorization Act

SECTIONS 302/304: Emergency Planning and Notification (40 CFR Part 355) Extremely Hazardous Substances: Ammonia is listed. Threshold Planning Quantity (TPQ): 500 lbs (227 kgs) Reportable Quantity (RQ): 100 lbs (45.4 kgs)

SECTIONS 311/312: Hazardous Chemical Reporting (40 CFR Part 370) IMMEDIATE HEALTH: Yes PRESSURE: Yes DELAYED HEALTH: No REACTIVITY: No FIRE: No
SECTION 313: Toxic Chemical Release Reporting (40 CFR Part 372) Ammonia is on the list of chemicals which may require reporting under Section 313.

CLEAN AIR ACT: SECTION 112 (r): Risk Management Programs for Chemical Accidental Release (40 CFR PART 68) Ammonia is listed as a regulated substance. Threshold Quantity (TQ): 10,000 lbs (4535 kgs)

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SECTION 313: Toxic Chemical Release Reporting (40 CFR Part 372) Ammonia is on the list of chemicals which may require reporting under Section 313.

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TSCA: Toxic Substance Control Act Ammonia is listed on the TSCA inventory. OSHA - OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION: 29 CFR Part 1910.119: Process Safety Management of Highly Hazardous Chemicals Ammonia is listed as a highly hazardous chemical. Threshold Quantity (TQ): 10,000 lbs (4535 kgs)

STATE REGULATIONS: CALIFORNIA: Accidental Release Prevention Program: Threshold Quantity (TQ): 100 lbs (45.4 kgs) Proposition 65: This product is not a listed substance which the State of California requires warning under this statute.

NEW JERSEY: Toxic Catastrophe Prevention Act: Registration Quantity (RQ): 5200 lbs (2358 kgs)
Placard: Non-Flammable EPA Regist No: None

Section 16	Other Information			
Flash Point (Test Method):	Non-Flammable		Flammable Limits	Lower Upper
Auto ignition Temperature:	850 deg C; 1560 deg F		(% by Volume)	16 25
Hazard Rating (N.F.P.A.):	Non-fire	Health: 2	Fire: 1 Reactivity: 0	Specific: Not-Applicable
	Fire:	Health: 3	Fire: 1 Reactivity: 0	Specific: Not-Applicable
	Liquefied:	Health: 3	Fire: 1 Reactivity: 0	Specific: Not-Applicable
	Gas:	Health: 2	Fire: 1 Reactivity: 0	Specific: Not-Applicable

ADDITIONAL HEALTH DATA Ammonia gas is very irritating to the mucous membrane of the nose, throat, and lungs, and high concentration can cause pulmonary edema. Irritation to the mucous membranes is noticeable at 100 ppm. Concentrations above 400 ppm will cause throat irritation and may destroy mucous surfaces upon prolong contact. Breathing air containing concentrations greater than 5000 ppm may cause sudden death from spasm or inflammation of the larynx. The following signs and symptoms result from exposure to gaseous ammonia. Burning of the eyes, conjunctivitis, skin irritation, swelling of the eyelids and lips, dry red mouth and tongue, burning in the throat, and coughing. In more severe cases of exposure, difficulty in breathing, sign and symptoms of lung congestion, and, ultimate death from respiratory failure due to pulmonary edema may occur.

ENVIRONMENTAL PROTECTION/IMPACT Certain geographical areas may have air pollution restrictions concerning use of materials in work situations which release volatile components to the atmosphere. Air pollution regulations should be studied to determine if this material is regulated in the area where it is to be used. In use and disposition of this material, care must be exercised to prevent contamination of water supplies. By increasing water alkalinity, it tends to diminish bacterial activity necessary for proper sewage. Concentrations as low as 0.3mg/l have been toxic to fish.

SPECIAL NOTES: Although anhydrous ammonia does not meet the Department of Transportation’s definition of flammable gas, under certain conditions and ignition temperatures, the gas may become flammable.

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